



TECHNICAL WHITEPAPER

Industrial LPDDR4X DRAM Chip

Ultra-low-power memory at 0.6 V I/O delivering up to 4266 MT/s in a compact dual-channel package, ideal for bandwidth-intensive yet power-sensitive designs.

JEDEC Compliant

Industrial & Extended Temp
–40°C to 85°C & –55°C to 105°C

2-8GB

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1. Executive Summary

The Loongtion LPDDR4X DRAM chip is a low-power, high-bandwidth memory solution designed for industrial, embedded, military-grade, and commercial systems requiring robust performance across extended temperature ranges. Based on traditional DDR SDRAM technology with architectural improvements in power efficiency and data throughput, this product utilizes wafer sourced from ChangXin Memory Technologies and is packaged to meet China-domestic supply compliance requirements.

Key differentiators include a dual-channel architecture per die, a Low Voltage Swing Terminated Logic (LVSTL) I/O system, and support for multiple temperature grades ranging from standard (-40°C to 85°C) and extended (85°C to 105°C) to industrial/military (-55°C to 105°C). The device operates at core voltages of VDD1 = 1.8 V, VDD2 = 1.1 V, and I/O voltage VDDQ = 0.6 V, enabling significant power savings while supporting data rates up to 4266 MT/s across all listed part numbers.

Loongtion LPDDR4X DRAM chip is suitable for applications requiring low power, high bandwidth, and supply-chain localization. All part numbers are offered in a compact FBGA200 package measuring 15 mm × 10 mm × 0.95 mm.

2. Product Overview

Loongtion LPDDR4X DRAM chip (Low Power Double Data Rate 4X Synchronous Dynamic Random Access Memory) is a high-performance memory product belonging to the LPDDR4X category. It is fabricated using advanced process technology and is delivered in an FBGA200 package with a 22-row ball grid array (ball pitch 0.8 mm along the X-axis and 0.65 mm along the Y-axis). The physical dimensions are 15 mm × 10 mm × 0.95 mm.

The device supports a multi-channel architecture: dual-channel die configurations with dedicated channels A and B, or functionally equivalent single-channel die stacking. The "_A" and "_B" signal suffixes denote respective DRAM channels; "_B" pads exist only on dual-channel DRAM devices.

Capacity and Temperature Options

Capacity	Part Number	Package	Data Rate (MT/s)	Temperature Range
2 GB	YMDB5CCBM-EA-A	FBGA200	3733 / 4266	-55°C to 105°C
4 GB	YMDB6CCBM-EA-A	FBGA200	3733 / 4266	-55°C to 105°C
2 GB	YZDB5CCBM-EA-A	FBGA200	3733 / 4266	-40°C to 85°C
4 GB	YZDB6CCBM-EA-A	FBGA200	3733 / 4266	-40°C to 85°C
4 GB	YZDB6CBAM-MA-B	FBGA200	3733 / 4266	-40°C to 85°C
8 GB	YZDB6CCDM-EA-A	FBGA200	3733 / 4266	-40°C to 85°C

Document Revision History

Revision	Date	Description
Rev 1.0	2024/7/16	Preliminary

Rev 2.0	2025/04/25	Added 8GB LPDDR4X product
Rev 3.0	2025/12/11	Added industrial/military-grade 2GB and 4GB product models
Rev 3.2	2026/03/02	Added 16-bit width

3. Technology and Architecture

Core Technology and Channel Architecture

The LPDDR4X DRAM chip is based on traditional DDR SDRAM technology with enhancements specifically targeting reduced power consumption and improved data throughput. Internal design is organized as a multi-channel architecture; for example, the 4 GB configuration comprises a dual-die, dual-channel, single-rank arrangement.

Each channel provides independent command/address and data buses. Differential clock inputs (CK_t_A/CK_c_A for channel A, CK_t_B/CK_c_B for channel B) and clock enable signals (CKE_A, CKE_B) control channel activation. The command/address inputs (CA[5:0]) are shared within each channel and operate in single data rate mode.

I/O System: LVSTL

The I/O system employs Low Voltage Swing Terminated Logic (LVSTL), which consists of a pull-up driver, a pull-down driver, and an on-die terminator. Calibration is performed in two steps:

1. **Pull-down calibration:** The NMOS pull-down device is calibrated to $240\ \Omega$ by connecting the ZQ pin through a $240\ \Omega \pm 1\%$ resistor to VDDQ. The drive strength is increased until the ZQ pin voltage is less than $VDDQ/2$.
2. **Pull-up calibration:** The NMOS pull-up device is calibrated to a VOH target value set via Mode Register Set (MRS). The ODT of the NMOS controller is configured, and drive strength increased until the data bit exceeds the VOH target.

Data Strobe and Data Mask Inversion

Each data byte has one pair of differential data strobe signals (DQS_t, DQS_c). During reads, the DRAM generates the strobe edge-aligned with data; during writes, the memory controller generates the strobe preceding data. The Data Mask Inversion (DMI) signal is bidirectional: it is driven high when data is inverted and low when normal. Data inversion can be disabled via the Mode Register. The DMI signal is also used for write data masking.

On-Die Termination (ODT)

For LPDDR4X devices, the ODT_CA pin is ignored. All ODT functions for CS, CA, and CK signals are fully controlled by Mode Registers MR11 and MR22. The ODT_CA pin should be connected to VDD2 or VSS.

Self-Refresh and Temperature Sensing

The device supports self-refresh with a 1× refresh rate before entering the high temperature range. An internal temperature sensor can be used to set the appropriate refresh rate and to determine whether AC timing derating is required.

Training and Calibration Features

Loongtion LPDDR4X DRAM chip includes a comprehensive set of training and calibration functions:

- **Bus Command Training (CBT)** – adjusts timing for command/address signals.
- **Write Leveling** – aligns DQS with clock for write operations.
- **Read Postamble Training** – optimizes read postamble timing.
- **DQS Internal Oscillator** – provides timing reference for DQS.
- **ZQ Calibration** – calibrates output drive strength and ODT.
- **Frequency Set Point (FSP)** – supports multiple operating frequencies.

Mode Register Operations

Mode register read and write operations are supported with the following minimum timing:

- **tMRR (Mode Register Read Cycle):** 8 nCK
- **tMRW (Mode Register Write Cycle):** MAX(10 ns, 10 nCK)
- **tMRD (Mode Register Set Command Delay):** MAX(14 ns, 10 nCK)

4. Key Features and Differentiators

- **Low Operating Voltages:** VDD1 = 1.8 V (1.70–1.95 V), VDD2 = 1.1 V (1.06–1.17 V), VDDQ = 0.6 V (0.57–0.65 V). The sub-1 V I/O supply contributes to reduced dynamic power consumption.
- **Dynamic Power Management and Deep Sleep Mode:** Enables further reduction in active and standby power.
- **Data Bus Inversion (DBI):** Supported with both DBI Off and DBI On modes; patterns for IDD4R and IDD4W are documented.
- **Multiple Temperature Grades:** Standard (-40°C to 85°C), Extended (85°C to 105°C), and Industrial/Military (-55°C to 105°C) available per part number.
- **China-Domestic Compliance:** Package design meets domestic supply requirements; corresponding China-domestic certification documents are available.
- **High Data Rate Support:** All listed part numbers support operation at 3733 MT/s and 4266 MT/s.
- **VREF Training:** Supports VREFCA and VrefDQ internal setting values obtained through training; Vref is set by the system for Ron and ODT settings.
- **1.1V High-Speed LVCMOS (HS_LVCMOS):** Supported for compatibility with common controller interfaces.

5. Technical Specifications

5.1 Package Dimensions (FBGA200)

Parameter	Symbol	Min (mm)	Nom (mm)	Max (mm)
Package height	A1	0.85	0.95	1.05
Standoff	A2	0.17	0.22	0.27
Package length	D	9.9	10.0	10.1
Die length	D1	—	8.8	—
Package width	E	14.90	15.00	15.10
Die width	E1	—	13.65	—
Ball pad width	d	—	0.80	—
Ball pitch (X-axis)	—	—	0.80	—
Ball pitch (Y-axis)	e	—	0.65	—
Ball diameter	b	0.26	0.31	0.36

5.2 Operating Voltage Ranges

Supply	Parameter	Min (V)	Typ (V)	Max (V)
VDD1	Core Voltage 1	1.70	1.80	1.95
VDD2	Core Voltage 2	1.06	1.10	1.17
VDDQ	I/O Supply Voltage	0.57	0.60	0.65

Note: VDD2 range of 1.06 V to 1.17 V applies to datasheet sections 9.3 and 9.4; VDDQ range of 0.57 V to 0.65 V applies to sections 9.5 and 9.6.

5.3 Absolute Maximum Ratings

Parameter	Min (V)	Max (V)
VDD1 Supply Voltage	-0.4	2.1
VDD2 Supply Voltage	-0.4	1.5
VDDQ Supply Voltage	-0.4	1.5
Other pins (VIN, VOUT)	-0.4	1.5

Warning: Stresses beyond absolute maximum ratings may cause permanent damage. Operation at maximum rating conditions for extended periods may affect device reliability.

5.4 Leakage Current

Parameter	Min	Max
Input Leakage Current (IL)	-4 μ A	4 μ A
Input/Output Leakage Current (IOZ)	-5 μ A	5 μ A

5.5 IDD Current Specifications (at 4266 Mbps, Dual-Channel, Single Die)

Parameter	VDD1 (mA)	VDD2 (mA)	VDDQ (mA)
IDD01	2.6	—	—
IDD02	—	41.5	—
IDD0Q	—	—	1.2

IDD4R2 (burst read)	—	284.2	—
IDD4RQ (burst read)	—	—	91.9
IDD4W2 (burst write)	—	235.2	—
IDD52 (all bank refresh burst)	—	105.4	—
IDD6 @25°C (self-refresh)	0.6	3.3	0.1
IDD6 @85°C (self-refresh)	0.8	5.2	0.1

Notes:

- IDD6 at 85°C is guaranteed; IDD6 at 45°C is typical (arithmetic mean distribution).
- IDD6ET (85°C to 105°C) is typical, sample value only, not tested.
- IDD parameters apply over full operating voltage and temperature ranges (except IDD6ET).
- Dual-channel devices are specified for dual-channel operation with both channels running simultaneously.
- Output load = 5 pF, RON = 40 Ω for IDD4R; ODT disabled: MR11[2:0]=000B.

5.6 Supported Data Rates

Data Rate (MT/s)	Notes
1600	Referenced in timing tables
1866	Referenced in timing tables
2133	Referenced in timing tables
2400	Referenced in timing tables
2667	Referenced in timing tables
3200	Referenced in timing tables
3733	Supported for all listed part numbers
4266	Supported for all listed part numbers

5.7 Clock Timing (at 4266 Mbps)

Parameter	Symbol	Min	Max
Average clock period	tCK(avg)	0.468 ns	100 ns

Note: Variation of tCK(avg) over 200 clock cycles can be up to $\pm 1\%$, provided all jitter and timing specifications are met. Average high pulse width (tCH(avg)) and average low pulse width (tCL(avg)) are defined in the datasheet; numerical min/max values are not specified in the source documentation.

5.8 Core Timing Parameters

Parameter	Symbol	Min	Max
CAS-to-CAS delay	tCCD	8 tCK(avg)	—
RAS to CAS Delay	tRCD	(18 ns, 4 nCK)	—
Row Precharge Time (Single Bank)	tRPpb	(18 ns, 4 nCK)	—
Row Precharge Time (All Banks)	tRPab	(21 ns, 4 nCK)	—
Row Active Time	tRAS	(42 ns, 3 nCK)	(9×tREFI, 70.2 μ s)
Write Recovery Time	tWR	(18 ns, 6 nCK)	—
Write-to-Read Delay	tWTR	(10 ns, 8 nCK)	—
Activate Bank A to Activate Bank B	tRRD	(10 ns, 4 nCK)	—

Precharge to Precharge Delay	tPPD	4 tCK	—
Four-Bank Activation Window	tFAW	40 ns	—

5.9 Read and Write Timing Parameters (Selected)

Parameter	Symbol	Min	Max
DQS output access time from CK_t/CK_c	tDQSCK	1.5 ns	3.5 ns
DQS-DQ skew	tDQSQ	—	0.18 UI
Write command to first DQS latching edge	tDQSS	0.75 tCK(avg)	1.25 tCK(avg)
Read preamble time	tRPRE	1.8 tCK(avg)	—
Read postamble time (0.5 tCK)	tRPST	0.4 tCK(avg)	—
Write postamble time	tWPRE	1.8 tCK(avg)	—

5.10 Temperature Ranges

Grade	Temperature Range	Applicable Part Numbers
Standard	-40°C to 85°C	YZDB5CCBM-EA-A, YZDB6CCBM-EA-A, YZDB6CBAM-MA-B, YZDB6CCDM-EA-A
Extended	85°C to 105°C	All standard-grade parts (derating applies)
Industrial/Military	-55°C to 105°C	YMDB5CCBM-EA-A, YMDB6CCBM-EA-A

Note: Operating temperature is defined as the case surface temperature at the center top of the LPDDR4X device.

5.11 Input/Output Capacitance (die only, at 3200–4266 MT/s)

Pin	Symbol	Min (pF)	Max (pF)
CK_t and CK_c	CCK	0.5	0.9
All other input-only pins	CI	0.5	0.9
DQ, DMI, DQS_t, DQS_c	CIO	0.7	1.3
ZQ pin	CZQ	0	5

Note: Capacitance values at 4266–3733 MT/s are not specified in source documentation (TBD).

5.12 Output Slew Rates

Parameter	Symbol	Min	Max
Single-ended output slew rate	SRQse	3 V/ns	9 V/ns
Output slew rate matching ratio	—	0.8	1.2
Differential output slew rate	SRQdiff	7 V/ns	18 V/ns

5.13 Input Voltage Levels (referenced to VDD2)

Parameter	Symbol	Min	Max
AC Input High Voltage	VIH(AC)	0.75×VDD2	VDD2+0.2 V
AC Input Low Voltage	VIL(AC)	-0.2 V	0.25×VDD2
DC Input High Voltage	VIH(DC)	0.65×VDD2	VDD2+0.2 V
DC Input Low Voltage	VIL(DC)	-0.2 V	0.2×VDD2

5.14 Overshoot/Undershoot Specifications

Parameter	Value
Maximum allowed peak amplitude in overshoot region	0.35 V
Maximum peak voltage allowed in undershoot region	0.35 V
Maximum overshoot region above VDD/VDDQ	0.8 V-ns

6. Performance and Reliability

Data Rate and Bandwidth

All listed part numbers support operation at 3733 MT/s and 4266 MT/s. The device provides a 16-bit data bus per channel (DQ[15:0]). Peak bandwidth per channel is not specified in source documentation; system designers should calculate based on the supported data rates and bus width.

Self-Refresh Current

Loongtion LPDDR4X DRAM chip exhibits very low self-refresh current, particularly at room temperature:

Temperature	VDD1 (mA)	VDD2 (mA)	VDDQ (mA)
25°C	0.6	3.3	0.1
85°C	0.8	5.2	0.1

Temperature Derating (Extended Range 85°C to 105°C)

When operating above 85°C, the following AC timing parameters must be derated:

Parameter	Derated Minimum
DQS output access time from CK_t/CK_c (tDQSCK)	3600 ps
RAS to CAS delay (tRCD)	tRCD + 1.875 ns
ACTIVATE to ACTIVATE command period (tRC)	tRC + 3.75 ns
Row active time (tRAS)	tRAS + 1.875 ns
Row precharge time (tRP)	tRP + 1.875 ns
Active group A to active group B (tRRD)	tRRD + 1.875 ns

Reliability Parameters

The following reliability parameters are not specified in source documentation: TBW (Terabytes Written), MTBF (Mean Time Between Failures), ECC (Error Correcting Code) support, and specific endurance ratings. Users should contact Loongtion for detailed reliability data.

Output Timing Characteristics

- **tDQSCK temperature variation:** 4 ps/°C
- **tDQSCK voltage variation:** 7 ps/mV
- **CK to DQS rank-to-rank delay:** ≤ 1 ns

7. Applications and Target Markets

Loongtion LPDDR4X DRAM chip is suitable for industrial and military-grade applications requiring low power, high bandwidth, and robust temperature tolerance. The device is available in standard (-40°C to 85°C) and industrial/military (-55°C to 105°C) temperature grades. Industrial and military-grade product models (2GB and 4GB) are available. Additional application details are not specified in the source documentation; contact Loongtion for specific use cases.

8. System Integration and Design Considerations

Pin Connection Requirements

- **ZQ pin:** Must be connected to VDDQ through a $240\ \Omega \pm 1\%$ resistor for calibration.
- **ODT_CA pin:** For LPDDR4X devices, this pin is ignored. It should be connected to VDD2 or VSS. ODT for CS/CA/CK is fully controlled via MR11 and MR22.

Power Supply Isolation

The three power supplies (VDD1, VDD2, VDDQ) are isolated from each other. Proper decoupling is required for each supply rail to meet the specified voltage tolerances.

Signal Integrity Requirements

- Differential clock and DQS signals must maintain monotonic slope through the input switching region.
- For differential clocks at 3733/4266 MT/s, the minimum differential input voltage is 360 mV, and the single-ended clock input voltage must be at least 180 mV.
- For DQS at 3733/4266 MT/s, the minimum differential input voltage is 340 mV.
- Overshoot and undershoot must be limited to 0.35 V peak amplitude and 0.8 V-ns overshoot region above VDD/VDDQ.

Receiver Mask Definitions

- The CA receiver mask defines the region where input signals must not intrude for correct capture. The center of the cumulative data input eye (V_{cent}) defines the reference level.
- The DQ receiver mask similarly defines valid input regions with total timing and voltage windows (T_{diVW_total} , V_{diVW_total}) for a target bit error rate.
- V_{ref} is set by the system for Ron and ODT settings.

Timing Derating for High Temperature

Designs operating at case temperatures above 85°C must apply the derating values listed in Section 6. The internal temperature sensor can be used to trigger refresh rate adjustments and to enable derating logic.

Self-Refresh Usage

The device supports self-refresh entry and exit with the following AC timing:

Parameter	Symbol	Min
Delay from SRE command to CKE input low	tESCKE	(1.75 ns, 3 tCK)
Minimum self-refresh time	tSR	(15 ns, 3 tCK)
Exit self-refresh to active command	tXSR	(tRFCab + 7.5 ns, 2 tCK)

Simulation Guidance

During circuit design, IBIS or other simulation tools should be used to correlate the timing reference load with the actual system environment.

9. Standards Compliance and Quality

China-Domestic Compliance

Loongtion LPDDR4X DRAM chip is designed and packaged to meet China-domestic supply requirements. Corresponding China-domestic certification documents are available upon request.

Wafer Source

Wafer production is sourced from ChangXin Memory Technologies.

Storage Temperature

The storage temperature range is -55°C to 125°C.

JEDEC Compliance

Specific JEDEC compliance status (e.g., JEDEC LPDDR4X standard) is not explicitly stated in the source documentation. The device follows conventional LPDDR4X electrical and timing parameters as described in this document.

Quality Parameters

Detailed quality parameters (ESD tolerance, RoHS compliance, reliability test data) are not specified in the source documentation. For current quality and compliance information, contact Loongtion at hi@loongtion.com.

10. Ordering Information

Capacity	Part Number	Package	Data Rate (MT/s)	Temperature Range	Grade
2 GB	YMDB5CCBM-EA-A	FBGA200	3733 / 4266	-55°C to 105°C	Industrial/Military
4 GB	YMDB6CCBM-EA-A	FBGA200	3733 / 4266	-55°C to 105°C	Industrial/Military
2 GB	YZDB5CCBM-EA-A	FBGA200	3733 / 4266	-40°C to 85°C	Standard
4 GB	YZDB6CCBM-EA-A	FBGA200	3733 / 4266	-40°C to 85°C	Standard
4 GB	YZDB6CBAM-MA-B	FBGA200	3733 / 4266	-40°C to 85°C	Standard
8 GB	YZDB6CCDM-EA-A	FBGA200	3733 / 4266	-40°C to 85°C	Standard

Notes:

- All part numbers support dual-channel operation.
- Additional ordering information (marking code, packaging options, tape/reel) is not specified in the source documentation. Contact Loongtion for details.

11. About Loongtion

Ningbo Loongtion Intelligent Technology Co., Ltd. (Loongtion®) is a supplier of China-domestic memory and solid-state storage products for industrial, embedded, medical, and commercial applications. The company's product portfolio includes:

- DDR3 / DDR4 / DDR5 / LPDDR4X DRAM chips
- eMMC 5.1 embedded storage
- M.2 NVMe SSDs
- NVMe BGA SSDs

Loongtion focuses on delivering reliable, temperature-robust memory solutions that meet the growing demand for localized supply chains in the Chinese market.

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Disclaimer: This whitepaper is based on the consolidated technical fact sheet provided by Ningbo Loongtion Intelligent Technology Co., Ltd. Specifications are subject to change without notice. Users should refer to the latest official datasheet for current parameters and ordering information. Loongtion assumes no liability for any errors or omissions in this document.